Experiment 8: Depth First Search (DFS) in Python

Aim

To write a Python program to implement the **Depth First Search (DFS)** algorithm.

Algorithm

1. Create a graph using an adjacency list.
2. Start from the source node.
3. Mark the source node as visited.
4. Visit each unvisited adjacent node recursively.
5. Continue until all nodes are visited.
6. Print the DFS traversal.

Program

```
graph = {
  'A': ['B', 'C'],
  'B': ['D', 'E'],
  'C': ['F'],
  'D': [],
  'E': [],
  'F': []
}

visited = []

def dfs(node):
  if node not in visited:
    print(node, end=" ")
    visited.append(node)
    for i in graph[node]:
      dfs(i)
```

dfs('A')

Input

Starting Node = A

Output

```
A B D E C F
```

Result

Thus, the Python program to implement the **Depth First Search (DFS)** algorithm was executed successfully, and the DFS traversal of the graph was displayed.